WORKED SOLUTION

STEP II 2018, Question 8

Pure • Bernoulli Differential Equations

STEP II 2018, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q8

The key idea

Both equations are Bernoulli equations: a power of y on the right spoils linearity, but the right power of y as a new variable makes the equation linear. With $y = v^2$ (or $y = v^3$) a common factor of v (or v^2) cancels and what remains is a first-order linear equation with constant coefficients, solved by an integrating factor or by separating variables. For the sketch, the neatest way to compare the two curves is to write both in terms of a single exponential, $u = e^{-t/6}$.

Solution

The question writes $y_1(x)$ and $y_2(x)$, but the independent variable is t throughout, so we write $y_1(t)$ and $y_2(t)$.

Part (i)

Put $y = v^2$ with $v = \sqrt{y} \geq 0$. Then $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 2v\dfrac{\mathrm{d}v}{\mathrm{d}t}$ and $y^{1/2} = v$, so the equation becomes

$$2v\frac{\mathrm{d}v}{\mathrm{d}t} = \alpha v - \beta v^2.$$

One solution is $v \equiv 0$, i.e. the constant solution $y \equiv 0$. Where $v \neq 0$ we can divide by v:

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{\beta}{2}v = \frac{\alpha}{2}.$$

This is linear with integrating factor $e^{\beta t/2}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{\beta t/2}\right) = \frac{\alpha}{2}e^{\beta t/2} \quad \Longrightarrow \quad ve^{\beta t/2} = \frac{\alpha}{\beta}e^{\beta t/2} + C.$$

So the general solution is

$$\sqrt{y} = \frac{\alpha}{\beta} + Ce^{-\beta t/2}, \qquad \text{i.e.} \qquad y = \left(\frac{\alpha}{\beta} + Ce^{-\beta t/2}\right)^2,$$

for a constant C (taken where the bracket is non-negative, since $v = \sqrt{y} \geq 0$), together with the constant solution $y \equiv 0$. Taking $C = 0$ gives the other constant solution $y \equiv \alpha^2/\beta^2$.

For $y_1(0) = 0$ we need $v(0) = 0$, so $C = -\alpha/\beta$, and

$$\boxed{y_1(t) = \frac{\alpha^2}{\beta^2}\left(1 - e^{-\beta t/2}\right)^2}$$

This is not constant, and the bracket $1 - e^{-\beta t/2}$ is non-negative for $t \geq 0$, so it is a genuine solution. (Dividing by v was valid for $t > 0$, where $v > 0$.)

Part (ii)

Now the power is $\frac{2}{3}$, so we want $y^{2/3}$ and y to share a factor. Put $y = v^3$, i.e. $v = y^{1/3} \geq 0$. Then $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 3v^2\dfrac{\mathrm{d}v}{\mathrm{d}t}$ and $y^{2/3} = v^2$, giving

$$3v^2\frac{\mathrm{d}v}{\mathrm{d}t} = \alpha v^2 - \beta v^3.$$

Ask yourself How would you guess the substitution $v = y^{1/3}$?

In part (i), $v = y^{1-\frac{1}{2}}$ made both sides share the factor v. For a Bernoulli equation $\dot{y} = \alpha y^n - \beta y$, the substitution $v = y^{1-n}$ always works, and here $1 - n = \frac{1}{3}$.

Apart from $v \equiv 0$ (the constant solution $y \equiv 0$), divide by v^2:

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{\beta}{3}v = \frac{\alpha}{3}.$$

With integrating factor $e^{\beta t/3}$, exactly as before,

$$y^{1/3} = \frac{\alpha}{\beta} + De^{-\beta t/3}, \qquad \text{i.e.} \qquad y = \left(\frac{\alpha}{\beta} + De^{-\beta t/3}\right)^3,$$

for a constant D (with the bracket non-negative). The condition $y_2(0) = 0$ forces $D = -\alpha/\beta$, so

$$\boxed{y_2(t) = \frac{\alpha^3}{\beta^3}\left(1 - e^{-\beta t/3}\right)^3}$$

Part (iii)

With $\alpha = \beta$,

$$y_1(t) = \left(1 - e^{-t/2}\right)^2, \qquad y_2(t) = \left(1 - e^{-t/3}\right)^3.$$

Shape of each curve. Both start at 0 and are increasing, since $1 - e^{-t/2}$ and $1 - e^{-t/3}$ increase from 0 towards 1. Both tend to 1 as $t \to \infty$, the constant solution $y = \alpha^2/\beta^2 = 1$

for (i) and $y = \alpha^3/\beta^3 = 1$ for (ii). Differentiating,

$$y_1' = e^{-t/2}\left(1 - e^{-t/2}\right), \qquad y_2' = e^{-t/3}\left(1 - e^{-t/3}\right)^2,$$

so both have zero gradient at the origin. The gradient $y_1' = w(1-w)$ with $w = e^{-t/2}$ is greatest at $w = \frac{1}{2}$, so y_1 has a point of inflection at $t = 2\ln 2$, $y_1 = \frac{1}{4}$. Similarly $y_2' = w(1-w)^2$ with $w = e^{-t/3}$ is greatest at $w = \frac{1}{3}$, giving an inflection at $t = 3\ln 3$, $y_2 = \frac{8}{27}$. Each curve is an S-shape rising from the origin to the asymptote $y = 1$.

Relative position. Let $u = e^{-t/6}$, so $0 < u < 1$ for $t > 0$, and $e^{-t/2} = u^3$, $e^{-t/3} = u^2$. Then

$$y_1 = (1-u^3)^2 = (1-u)^2(1+u+u^2)^2, \qquad y_2 = (1-u^2)^3 = (1-u)^3(1+u)^3,$$

and so

$$\frac{y_1}{y_2} = \frac{(1+u+u^2)^2}{(1-u)(1+u)^3}.$$

Expanding, $(1+u+u^2)^2 = 1+2u+3u^2+2u^3+u^4$ and $(1-u)(1+u)^3 = (1-u^2)(1+u)^2 = 1+2u-2u^3-u^4$. The numerator minus the denominator is

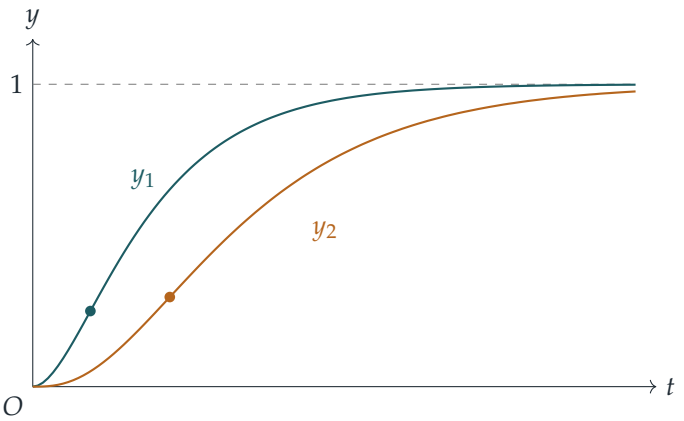
$$3u^2 + 4u^3 + 2u^4 > 0 \qquad (0 < u < 1),$$

and the denominator is positive, so $y_1/y_2 > 1$. Hence

$$y_1(t) > y_2(t) \quad \text{for all } t > 0:$$

the curves meet only at the origin, and y_1 lies above y_2 everywhere else.

This agrees with the behaviour at both ends. Near $t = 0$, $y_1 \approx \frac{t^2}{4}$ while $y_2 \approx \frac{t^3}{27}$, which is much smaller. For large t, $1 - y_1 \approx 2e^{-t/2}$ and $1 - y_2 \approx 3e^{-t/3}$, and $e^{-t/3}$ decays more slowly, so y_2 is further below the asymptote.



The dots mark the points of inflection, $(2\ln 2, \frac{1}{4})$ on y_1 and $(3\ln 3, \frac{8}{27})$ on y_2.

Checks

- Substitute back: for y_1 with $\alpha = \beta = 1$, $y_1^{1/2} - y_1 = (1-w) - (1-w)^2 = w(1-w)$ with $w = e^{-t/2}$, which is exactly y_1'.
- Large t: $y_1 \to \alpha^2/\beta^2$, the non-zero point where $\alpha y^{1/2} = \beta y$; likewise $y_2 \to \alpha^3/\beta^3$, where $\alpha y^{2/3} = \beta y$.
- Numbers with $\alpha = \beta = 1$: at $t = 2$, $y_1 \approx 0.400$ and $y_2 \approx 0.115$; at $t = 10$, $y_1 \approx 0.987$ and $y_2 \approx 0.897$. Both formulas were also confirmed by a computer step-by-step integration of the differential equations.

Common slips

- Dividing by v (or v^2) without noting that $y \equiv 0$ is also a solution. That is why the question asks for the *non-constant* solution through the origin.
- Getting the exponent wrong: the substitution $y = v^3$ gives $e^{-\beta t/3}$, not $e^{-\beta t/2}$.
- Arguing the order of the curves from the gradients alone. Where the curves have the same height $y \in (0,1)$, $y^{2/3} > y^{1/2}$, so y_2 would be rising faster there; this does not say which curve is higher, so a direct comparison is needed.

Beyond STEP

These are Bernoulli equations, a standard first-year family solved by $v = y^{1-n}$. They also illustrate non-uniqueness: since $y^{1/2}$ and $y^{2/3}$ have infinite gradient at $y = 0$, the right-hand side is not Lipschitz there, and $y \equiv 0$ and y_1 both pass through the origin. In fact, staying at 0 until any time t_0 and then following $y_1(t - t_0)$ gives a whole family of solutions. The Picard–Lindelöf theorem explains exactly when this cannot happen. Boyce and DiPrima, *Elementary Differential Equations*, covers both Bernoulli equations and existence and uniqueness.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.